NACS 645 – The origins of social cognition

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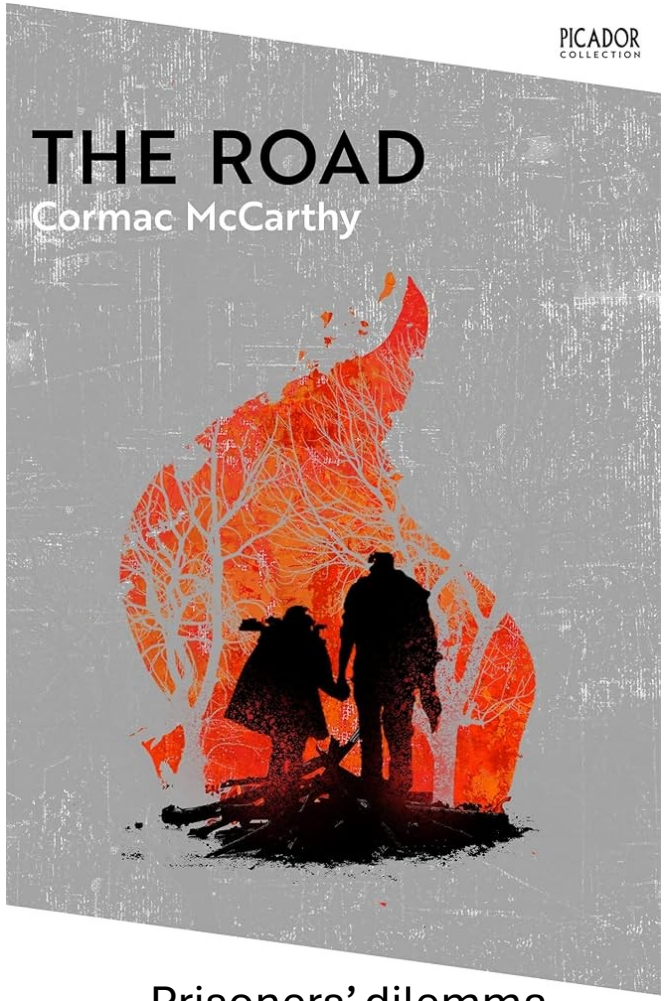
Choosing actions – Game theory



Game Theory assumptions:

- Each agent has its own description of states of the world (beliefs)
- Each agent has a utility function (preferences, uncertainty profile)
- Each agent maximizes expected utility (decision-theoretic rationality)

Given these assumptions, each player ***best responds*** to the others (*Nash equilibrium*)

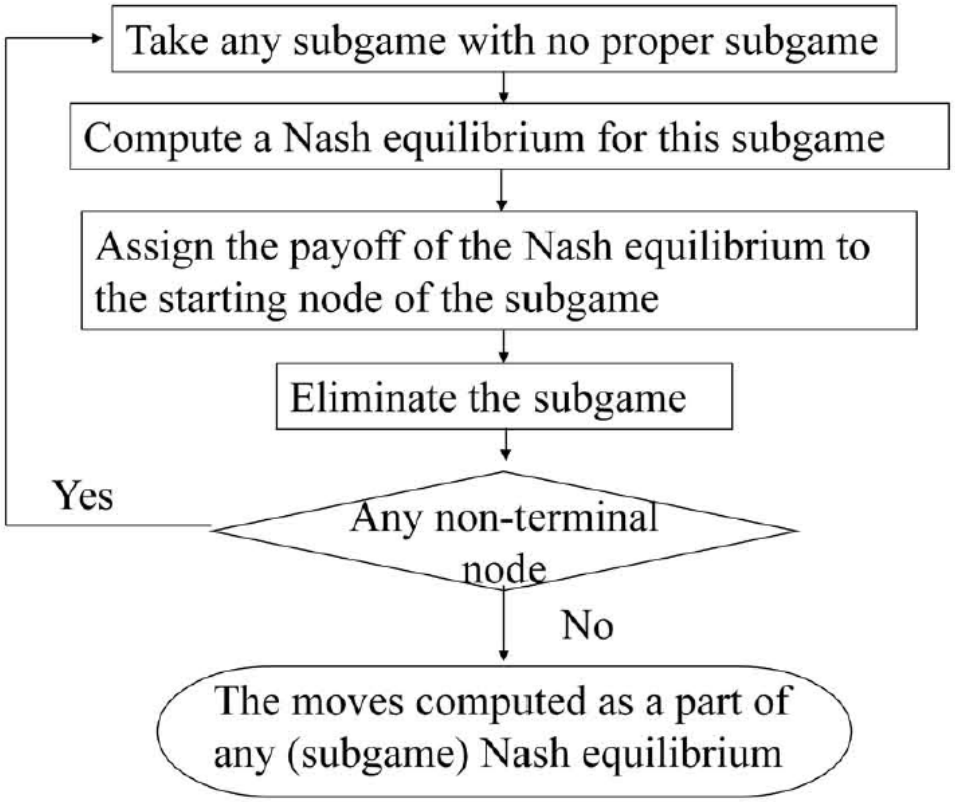
Nash equilibrium:

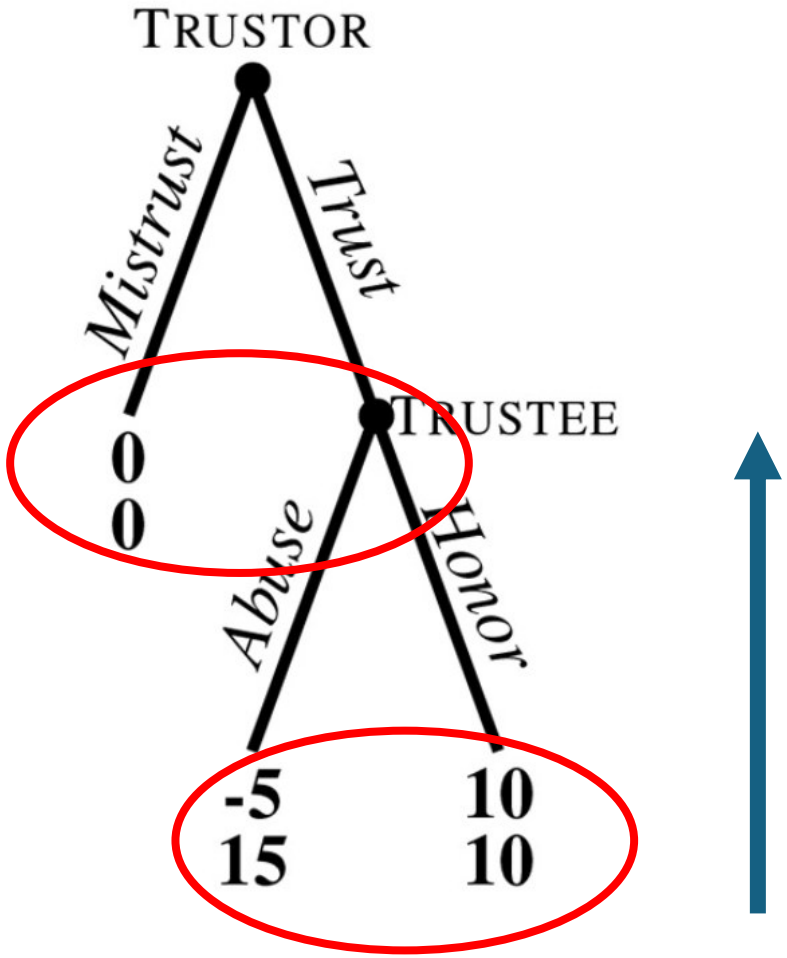
- A set of actions (action profible that is stable) with which each player's action maximizes his/her payoff given the actions of the others
- Each player's strategy is optimal given the strategies of all other players, so no player can benefit by unilaterally changing their action.
- Over time, as players learn the game, equilibria tend to emerge, while non-equilibrium profiles should disappear.

Prisoners' dilemma

	C	*D*
C	-1,-1	-4,0
D	0,-4	-3,-3

How to find the best action – backward induction





If we can prevent the defection at the last trial n (and all n_{-i} trials), we can incentivize cooperation (e.g., infinitely and indefinitely repetead games)

Game Theory and Nuclear deterrence

The TSEconomist Student Magazine

Game Theory & Nuclear Strategy

Game theory tends not to be the most popular class for many M1 Economics students. Mathematical formalism and rigorous logical arguments tend to scare or bore rather than engage. What makes game theory truly exciting, however, is its wide range of applications from evolution biology to political science. The discussion of military strategy during the Cold War was the crucial catalyst that brought game theory onto the stage in the first place. This article will give a historical overview of the development of nuclear strategy during the Cold War era and show parallels to game theoretic applications.

Philip Hanspach, 2016, Game Theory & Nuclear Strategy, available at: *The Tseconomist*

How To Win A Nuclear Standoff

President Trump and Kim Jong Un's saber-rattling is dangerous, but not irrational

By Oliver Roeder
Filed under The Trump Administration
Published Sep. 6, 2017

Oliver Roeder, 2017, How to win a nuclear standoff, available at: *FiveThirtyEight*

Pure and mixed games

Stag Hunt

	Stag	*Hare*
Stag	8,8	0,7
Hare	7,0	5,5

1 Nash equilibrium

Side of the road

	Left	*Right*
Left	1,1	0,0
Right	0,0	1,1

2 Nash equilibria

Pure coordination

Prisoners' dilemma

	C	*D*
C	-1,-1	-4,0
D	0,-4	-3,-3

1 Nash equilibrium

Battle of the sexes

	B	*F*
B	2,1	0,0
F	0,0	1,2

2 Nash equilibria

Mixed motive: coop / competition

Matching pennies

	Heads	*Tails*
Heads	1,-1	-1,1
Tails	-1,1	1,-1

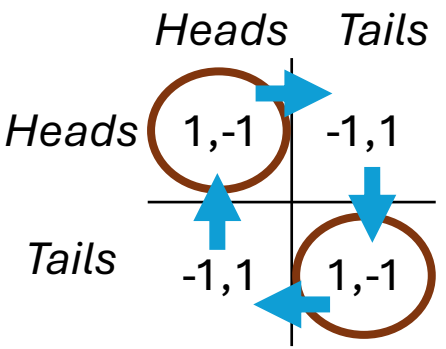
0 Nash equilibrium

Pure competition

Mixed strategies

- In **strategic environments with mixed or competitive motives, deterministic strategies are exploitable**:
  predictability creates information asymmetry that can be exploited
- Idea:
  - a) confuse the opponent by introducing randomness,
  - b) avoiding patterns by randomizing completely
- Let's define a strategy (s_i) for the agent (i) as any probability distribution over the set of actions (A_i)
  - Pure strategy: only one action played with $p = 1$
  - Mixed strategy: more than one action played with $p > 0$
- In matching pennies, optimal strategy is to play each action a_i with equal proba $p = \frac{1}{|A_i|}$

Matching pennies



0 Nash equilibrium

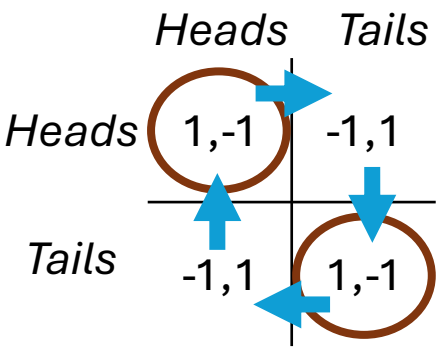
Pure competition

Mixed strategies – exploiting others

Based on the works of Matthew O. Jackson, Kevin Leyton-Brown & Yoav Shoham, Game Theory, on *Stanford online*.

- In matching pennies, optimal strategy is to play each action a_i with equal proba $p = \frac{1}{|A_i|}$
  - Any player deviating from $p = \frac{1}{|A_i|}$ becomes exploitable:
    if P2 plays H,T with [.55,.45], P1 best response is H with $\mathrm{p} = 1$
  - Any player using $p = \frac{1}{|A_i|}$ is unexploitable:
    [.45,.55] vs [.5,.5] would yield utility $u_1 = u_2 = 0$
- Yet, humans struggle to generate truly random sequences, making them exploitable

Matching pennies



0 Nash equilibrium

Pure competition

- Players may instead rely on pseudo-random or deceptive strategies, faking patterns to induce false beliefs in opponents. Efficacy depends on types of games and opponents:
  - It can succeed only if player remains unpredictable, e.g., due to uncertainties
  - GTO players are guaranteed immunity from exploitation and may detect patterns
  - Non-GTO players can overfit perceived regularities

Interpreting mixed strategies

Matching pennies

	Heads	*Tails*
Heads	1,-1	-1,1
Tails	-1,1	1,-1

Battle of the sexes

	B	*F*
B	2,1	0,0
F	0,0	1,2

Different interpretations of mixed strategies

- Randomize to confuse the opponent
  (e.g., matching pennies)
- Randomize when uncertain about the other's action
  (e.g., battle of the sexes)
- Concise description of what happens in repeated play
  (probability distributions over sets of actions)
- Describes population dynamics
  (e.g., 2 agents drawn from a population, all with deterministic strategies. Mixed strategies gives the probability of getting each pure strategy available)

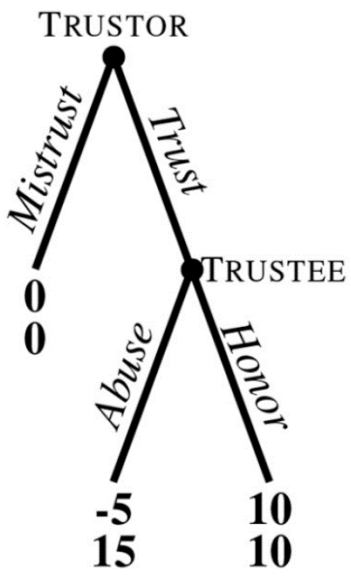
Perfect vs Imperfect information games

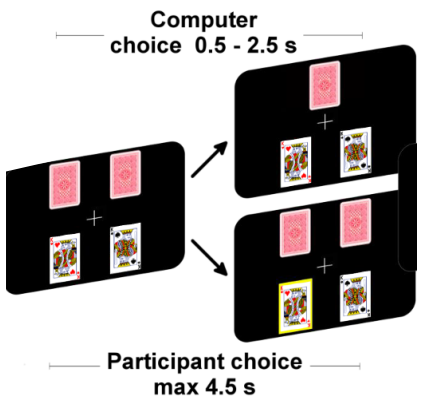
Perfect information games

When all players have access to all the information of a game.
(e.g., Trust game, Chess, Tic-tac-toe)

Easy to solve:

- Every player knows all past actions and the full state of the game
- Backward induction ensures a pure strategy Nash equilibrium
- Randomization is unnecessary, because decision is transparent





Imperfect information games

When some actions of states are hidden from at least one player
(e.g., MP, Rock-Paper-Scissors, Poker)

Harder to solve:

- Players do not observe some past moves or states (e.g., simultaneous moves, hidden cards)
- Players' may better be probabilistic
- Pure-strategy equilibria may not exist, but mixed-strategy equilibria do

Incomplete information games

What if players are **uncertain about the *structure* of the game** itself?
→ number of players, available actions, payoffs for each action profile

- Each player faces several *possible games* they *believe* could be the real one
- All players share a **common prior** over possible games, but receive **private signals** to **update their beliefs about the true game**

Interpretation**:**
- Some players know information that others do not
- Each player's *epistemic type* encodes *what they know* and *what they believe others know*
- **Optimal actions depend on beliefs about others'** epistemic types

- Deception manipulates others' beliefs, exploiting information asymmetries
- Predicting and interpreting others' moves requires modeling their *beliefs*, not just their *payoffs – i.e., mentalizing*

	$I_{2,1}$	$I_{2,2}$
$I_{1,1}$	MP: $\begin{pmatrix} 2,0 & 0,2 \\ 0,2 & 2,0 \end{pmatrix}$, $p = 0.3$	PD: $\begin{pmatrix} 2,2 & 0,3 \\ 3,0 & 1,1 \end{pmatrix}$, $p = 0.1$
$I_{1,2}$	Coord: $\begin{pmatrix} 2,2 & 0,0 \\ 0,0 & 1,1 \end{pmatrix}$, $p = 0.2$	BoS: $\begin{pmatrix} 2,1 & 0,0 \\ 0,0 & 1,2 \end{pmatrix}$, $p = 0.4$

Incomplete information games

Task setup

- There are **4 possible games**: *Matching Pennies (MP), Prisoner's Dilemma (PD), Coordination (Coord),* and *Battle of the Sexes (BoS)*
- **Nature** randomly selects which one is played (not publicly revealed)
- Each player receive a private signal

Information structure: let's say all players received private info

- **P1** knows they are in the **bottom row** → game is either *Coord* or *BoS*
- **P2** knows they are in the **right column** → the game is either *PD* or *BoS*

Interpretation

- Each player faces **uncertainty about the true game** and must reason about the opponent's beliefs and likely behavior
- Both share a **common prior** over which of the four games Nature chose

Inference logic

- If game is *Coord*, P2 will play *MP* or *Coord*; otherwise P2 will play *PD* or *BoS*. By observing P2, P1 can update beliefs about the true game
- **e.g., if P2 doesn't play pure strategy, the game is *BoS***

	$I_{2,1}$	$I_{2,2}$
$I_{1,1}$	MP: $\begin{bmatrix} 2,0 & 0,2 \\ 0,2 & 2,0 \end{bmatrix}$ $p = 0.3$	PD: $\begin{bmatrix} 2,2 & 0,3 \\ 3,0 & 1,1 \end{bmatrix}$ $p = 0.1$
$I_{1,2}$	Coord: $\begin{bmatrix} 2,2 & 0,0 \\ 0,0 & 1,1 \end{bmatrix}$ $p = 0.2$	BoS: $\begin{bmatrix} 2,1 & 0,0 \\ 0,0 & 1,2 \end{bmatrix}$ $p = 0.4$

Mentalizing, central to strategic interactions

Tomasello, 2008. *MIT Press*.
Tomasello, 2020. *Episteme*.
Sperber et al., 2010. *Mind & Language*.

Cooperative communication

- Aligning beliefs and actions requires mutual **transparency** of minds
- Language evolved for coordination and shared understanding (Tomasello, 2008, 2020): sharing mental states, referencing *what is*, building ground truth in common reality
- But transparency allows **exploitation**: free riders gain if they remain undetected

Need for truthfulness

- Stable cooperation requires a) most communication be **truthful**; and b) truth is the default expectation
- Yet, this enables strategic deception and exploiting others
- Too many free riders → collapse of cooperation

Dual pressures

- Mentalizing makes cooperation possible and deception feasible
- Lying is effortful because it requires simulating others' minds
- Communication evolved under **dual pressures**: **cooperation** through truth-sharing **vs. competition** through manipulation

Epistemic vigilance

- To protect cooperation, evolution favored mechanisms for detecting dishonesty (Sperber, 2010)
- Cognitive systems that assess reliability and sincerity of communicated information, acting as counterweights to gullibility

The development of ToM

Bettle & Rosati, 2021. *Language learning and Development.*
Fujita, Devine, Hughes, 2022. *Cognitive Development.*
Rakoczy, 2022. *Nature Reviews Psychology.*

(shared with other primates)		*(human-unique?)*	**function**
Simple gaze following	→	Joint attention; sustained attention to others	• Recognize positive intentions • Share attention with others
Goal and intention attribution	→	Attributing cooperative or shared intentions	• Identify cooperative opportunities • Understand helpful communication • Maintenance so all parties benefit
Level 1 Perspective taking (presence/absence)	→	Level 2 perspective-taking ('how' it looks)	• 'Bird's-eye view' of joint acts • Facilitate coordination
Attribute knowledge and ignorance	→	Attribute false beliefs	• Facilitate coordination • Avoid miscommunication • Reputation management

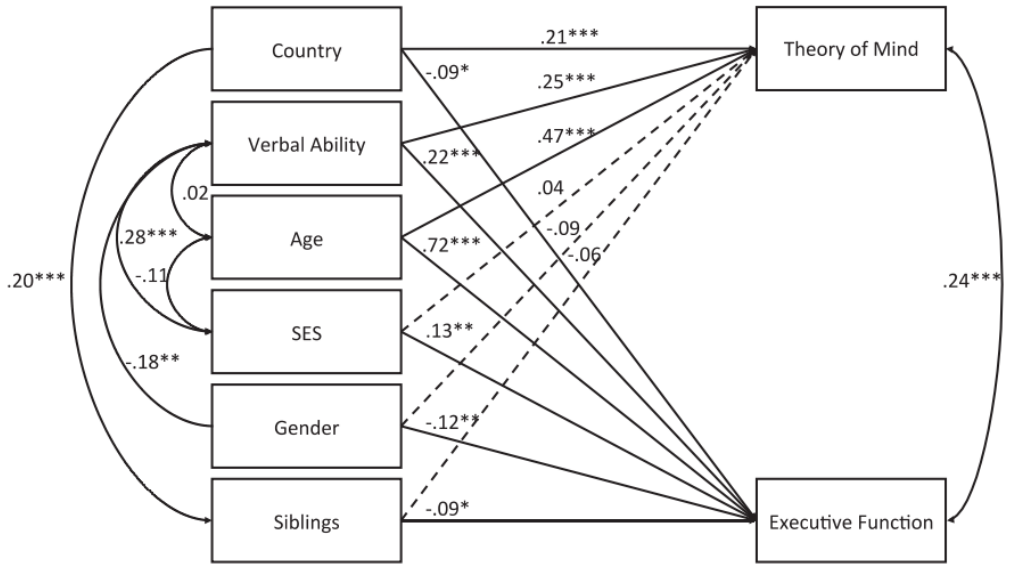
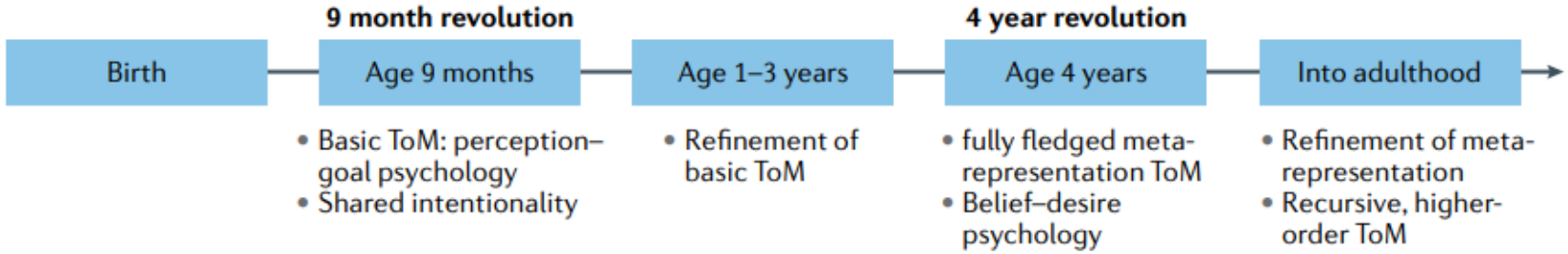



Fig. 1. A structural equation model of cross-cultural differences in theory of mind and executive function with covariates. Country: 0 = Japan, 1 = UK. Gender: 0 = girls, 1 = boys. SES = Socioeconomic Status. *Note.* * $p < .05$. ** $p < .01$. *** $p < .001$.